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END SEMESTER EXAMINATION 2025

Name of the Course: B.Tech Petroleum Engineering

Semester: VI

Name of the Paper: Natural Gas Engineering

Course Code: TPE-603

Time: Three Hours

MM: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions in each main question.
- iii. Total marks for each main question is **twenty**
- iv. Each Question carries **10 marks**

Q1		COs												
a.	Define the following: i. Retrograde condensation ii. Critical point iii. Bubble point iv. Dew point v. Gas hydrates	CO1, CO2												
b.	Define the quality lines in phase diagram? Draw the PT and PV phase diagram for single component system showing the different region What is natural gas? Discuss all the properties with mathematical expressions of Natural gas.	CO1, CO2												
c.	Calculate the gas specific gravity of natural gas in the reservoir having the following composition. <table border="1"><thead><tr><th>Composition</th><th>Mole fractions</th></tr></thead><tbody><tr><td>C1</td><td>0.80</td></tr><tr><td>C2</td><td>0.081</td></tr><tr><td>C3</td><td>0.020</td></tr><tr><td>CO2</td><td>0.030</td></tr><tr><td>H2S</td><td>0.015</td></tr></tbody></table> The reservoir is having at initial condition of $P_i = 3500\text{psia}$ and 185°F. By calculating Sp. Gr. Of the gas also calculate T_{pc}, P_{pc}, T_{pr}, P_{pr} by using Standing and Katz correlations (take compressibility factor $Z = 0.87$ for the data given) and hence the density of natural gas. Compare the density of the natural gas in case of ideal gas.	Composition	Mole fractions	C1	0.80	C2	0.081	C3	0.020	CO2	0.030	H2S	0.015	CO3
Composition	Mole fractions													
C1	0.80													
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Q2														
a.	With the help of diagram explain the extraction of natural gas by the enhanced (CBM) techniques from CBM? Write down the usage and advantages of CBM. What do you mean by proppant? What are the functions of proppant and discuss the types of proppant?	CO3, CO4												

b.	Describe the steps involved in the extraction of shale gas? Why hydraulic fracturing needed in the production of shale gas? Explain hydraulic fracturing? Illustrate the different theory and techniques for the hydraulic fracturing?	CO1, CO4
c.	How gas hydrates formed? Discuss the various issues related to extraction of natural gas hydrates? Gas hydrates extraction requires the unconventional approach why? Explain the different methods in the extraction of gas hydrate	CO1, CO2
Q3		
a.	Discuss Dehydration of Natural gas in details? What are the effects of presence of water in natural gas?	CO1
b.	What are the information's that is to be known for the design of Glycol dehydrator? In the design of Glycol dehydrator explain glycol contractor, water rate and required glycol circulation rate along with their mathematical expression.	CO1, CO2
c.	What are the dehydration systems? Discuss absorption and adsorption process in the dehydration of natural gas	CO1, CO5
Q4		
a.	Explain the following along with mathematical expression in the determination of water content of natural gas: i. The gravity correction factor ii. The salinity correction factor iii. Mole percent of H ₂ S equivalent concentration iv. Can water content be calculated by the dew point?	CO1, CO5
b.	Define recording charts? What are the two types of principal chart used? Discuss in details about direct reading charts.	CO4, CO5
c.	What are the different methods/instruments used in the volumetric measurement of natural gas. Discuss orifice meter for the measurement of natural gas with required equations used	CO3, CO6
Q5		
a.	Discuss natural gas utilization as fertilizer and petrochemical feed stock in details?	CO4, CO6
b.	Calculate the maximum quantity of gas that can be transported through 4" dia, 20 Km long horizontal pipeline under the following conditions : Inlet pressure = 2100 psia Maximum pressure drop = 550 psi Average gas temp. = 100 F Pipeline efficiency factor = 0.92 Gas gravity = 0.6 Z _{avg} can be taken as = 0.9 Use both Panhandle B and Weymouth methods	CO4, CO6
c.	Define transmission factor? Tabulate and discuss the different transmission factor with different flow equations	CO2, CO3